Example Three, Variable Capacity – Assume that two different types of haul units are available for the project described in Example Two. The paving company owns haul units that carry 24 tonnes (26.5 tons). There will be eight of those haul units assigned to the project. Other haul units will be hired, but they transport an average of 20 tonnes (22 tons). How many of the smaller haul units are needed?

First, calculate how many tonnes the eight larger trucks can transport when they all complete a round trip and then divide by the cycle time.

8 trucks x 24 tonnes ÷ 1.5 hours = 128 tonnes/hour

Next, subtract the weight carried by the larger trucks from the total hourly production.

225 tonnes minus 128 tonnes = 97 tonnes per hour remaining

TRUCKING CALCULATOR (LARGE UNITS)

Trucking	**General Inputs**		*Imperial Units*	*Metric Units*
Paver Speed	Production Rate of Hot Plant:		[141] *tons/hr*	[128] *tonnes/hr*
Compaction	Multiple Silo Plants: Initial Storage		[1] *tons*	[0] *tonnes*
Windrow	Paving Hours:		[0.0] *hrs*	[0.0] *hrs*
Yield	Truck Capacity (size):		[26.5] *net tons*	[24.0] *net tonnes*
Slope	**Truck Cycle Times** *(minutes)*			
Thickness	Load Time and Ticket:	[6]		
Job Summary	Tarp:	[4]		
Legal	Haul to Job:	[35]		
	Time on Site:	[2]		
	Dump / Clean:	[8]		
	Return Haul:	[35]		
Exit	**Truck Cycle Factor** *(total time in hours)*	[1.5]		
	Number of Trucks Needed:	[8.0]		

TRUCKING CALCULATOR (SMALL UNITS)

Trucking	**General Inputs**		*Imperial Units*	*Metric Units*
Paver Speed	Production Rate of Hot Plant:		[107] *tons/hr*	[97] *tonnes/hr*
Compaction	Multiple Silo Plants: Initial Storage		[0] *tons*	[0] *tonnes*
Windrow	Paving Hours:		[0.0] *hrs*	[0.0] *hrs*
Yield	Truck Capacity (size):		[22.0] *net tons*	[20.0] *net tonnes*
Slope	**Truck Cycle Times** *(minutes)*			
Thickness	Load Time and Ticket:	[6]		
Job Summary	Tarp:	[4]		
Legal	Haul to Job:	[35]		
	Time on Site:	[2]		
	Dump / Clean:	[8]		
	Return Haul:	[35]		
Exit	**Truck Cycle Factor** *(total time in hours)*	[1.5]		
	Number of Trucks Needed:	[7.3]		

Example Four, variable traffic – In the previous examples, the cycle time—1.5 hours—has been constant. Now, assume that the haul units (all carrying 24 tonnes / 26.5 tons) will be traveling through peak traffic for two hours at the start of the shift and for two hours at the end of the shift. During peak traffic hours, the cycle time will increase to 2.2 hours.

TRUCKING CALCULATOR

Trucking	**General Inputs**	*Imperial Units*	*Metric Units*	
Paver Speed	Production Rate of Hot Plant:	[220]	[200]	*tons/hr* / *tonnes/hr*
Compaction	Multiple Silo Plants: Initial Storage	[276]	[250]	*tons* / *tonnes*
Windrow	Paving Hours:	[10.0]	[10.0]	*hrs* / *hrs*
Yield	Truck Capacity (size):	[26.5]	[24.0]	*net tons* / *net tonnes*
Slope	**Truck Cycle Times** *(minutes)*			
Thickness	Load Time and Ticket:	[6]		
Job Summary	Tarp:	[4]		
Legal	Haul to Job:	[55]		
	Time on Site:	[2]		
	Dump / Clean:	[8]		
	Return Haul:	[55]		
Exit	**Truck Cycle Factor** *(total time in hours)*	[2.2]		
	Number of Trucks Needed:	[20.6]		

[CALCULATING PAVING SPEED]

Calculating the correct paving speed is necessary to ensure efficient paving—one of the basic principles of quality paving. Unless the project has many short pulls and lots of areas where the crew will be doing handwork, the paving speed should be established to develop at least 75 percent efficiency. That is to say, the paver is laying down mix at least 45 minutes out of every hour and stopped for truck exchanges or waiting for mix no more than 15 minutes out of every hour. On a highway project with long pulls and using a transfer vehicle, the paver can approach 100 percent efficiency with careful planning.

Continuous paving at a constant speed is the goal when smoothness is desired. Uniform layer temperatures are also more likely when the paver makes only a few short stops during the shift.

Calculate the paving speed, communicate the paving speed, and stay with the paving speed unless something changes to affect the hourly production rate. Normally, the paving superintendent or the crew supervisor calculates the paving speed and tells the paver operator what speed to maintain.

Paving speed can be calculated if the following information is known:

- Planned hourly output of the asphalt manufacturing facility, including storage bins
- Paving width
- Paving thickness (loose or compacted)
- Weight of the material (loose or compacted). The weight of the material can be obtained from the operator of the asphalt manufacturing facility or from the quality control staff.

To calculate the paving speed manually:

Metric Units

$$\text{tons / hour} \times \frac{1000\ \text{kg}}{60\ \text{min}} (16.67) \div \text{width (m)} \div \text{depth (m)} \div \text{kg / m}^3 = \text{m / min}$$

The paving speed that results from this formula is at 100% efficiency. The paving speed may have to be multiplied by an efficiency factor to account for truck exchanges and stops waiting for mix.

For example, an 85% efficiency factor may be used. The calculated paving speed would then be multiplied by 0.85.

Imperial Units

$$\text{tons / hour} \times \frac{2000\ \text{lb}}{60\ \text{min}} (33.33) \div \text{width (ft)} \div \text{depth (ft)} \div \text{lb / ft}^3 = \text{ft / min}$$

The second page of the Paving Production Calculator provides a faster way of calculating paving speed than doing a manual computation.

Again, the Paving Production Calculator is available through Cat dealers or as an App from the appropriate App stores.

Example One – On this project, the hourly production rate is 200 tonnes (220 tons) per hour. The uncompacted paving depth is 75 mm (3").

The paving width is 5.5 m (18'). The weight of the uncompacted material is 2162 kg/ m³ (135 lb/ft³).

PAVER SPEED CALCULATOR

Trucking
Paver Speed
Compaction
Windrow
Yield
Slope
Thickness
Job Summary
Legal

Exit

General Inputs	*Imperial Units*	*Metric Units*
Paving Thickness:	[2.95] *in*	[75] *mm*
Paving Width:	[18.00] *feet*	[5.486] *meter*
Material Density Uncompacted:	[135] *lbs/ft³*	[2162] *kg/m³*
Paver Speed @ Given Production Rate		
Production Rate of Hot Plant:	[220] *tons/hr*	[200] *tonnes/hr*
Calculated Paving Speed - 100% Efficiency:	[12.3] *ft/min*	[3.75] *m/min*
Calculated Paving Speed - 95% Efficiency:	[12.9] *ft/min*	[3.94] *m/min*
Calculated Paving Speed - 90% Efficiency:	[13.5] *ft/min*	[4.12] *m/min*
Calculated Paving Speed - 85% Efficiency:	[14.1] *ft/min*	[4.31] *m/min*
Calculated Paving Speed - 80% Efficiency:	[14.8] *ft/min*	[4.50] *m/min*
Calculated Paving Speed - 75% Efficiency:	[15.4] *ft/min*	[4.69] *m/min*
Effective Paving Speed:	[12.3] *ft/min*	[3.75] *m/min*

The Paver Speed Calculator provides paving speeds based on six efficiency factors.

100% efficiency: 3.75 m/min (12.3'/min) – Theoretically, using a material transfer vehicle, the paver can pave continuously without stopping or changing the paving speed. It is usually unrealistic to expect the paver to work at one continuous speed for an entire shift. Events happen to cause slight delays in trucking, for example. So, normally, a lower efficiency is selected.

95% efficiency: 3.94 m/min (12.9'/min) – If a 95% efficiency factor is selected, the actual working speed will be slightly higher than the actual speed at 100% efficiency. The higher working speed is used to make up for a few short stops during that hour. The effective speed is still 3.75 m/min (12.3'/min).

90% efficiency factor: 4.12 m/min (13.5'/min) – Planning for lower efficiency factors, the actual working speed continues to increase little by little while the effective paving speed stays the same. Realistically, a 90% to 95% efficiency factor would probably be selected when using a material transfer device.

85% efficiency factor: 4.31 m/min (14.1'/min) – If material is transferred directly from haul units to the hopper of the paver, the efficiency factor selected will continue to decrease. The approximated maximum that can be achieved without a transfer device is 85%.

80% efficiency factor: 4.5 m/min (14.8'/min) – Lower efficiency factor equals higher actual speed. The effective paving speed remains the same.

75% efficiency factor: 4.7 m/min (15.4'/min) – In general, on projects that have mostly long pulls with the paver, the minimum efficiency factor that is consistent with quality results is 75%. If less than 75% efficient, there are too many long paver stops. The paving speed will have to be slowed to reduce the amount of time the paver is stopped. Or, perhaps, a way can be found to shorten the truck exchange time.

Example Two – Assume that all factors remain the same as in Example One, except that the paving width is now 3.7 m (12'). Enter the factors into the Paving Production Calculator.

PAVER SPEED CALCULATOR

Trucking	**General Inputs**	*Imperial Units*	*Metric Units*
Paver Speed	Paving Thickness:	[2.95] *in*	[74.9] *mm*
Compaction	Paving Width:	[12.00] *feet*	[3.658] *meter*
Windrow	Material Density Uncompacted:	[135] *lbs/ft³*	[2162] *kg/m³*
Yield	**Paver Speed @ Given Production Rate**		
Slope	Production Rate of Hot Plant:	[220] *tons/hr*	[200] *tonnes/hr*
Thickness	Calculated Paving Speed - 100% Efficiency:	[18.4] *ft/min*	[5.61] *m/min*
Job Summary	Calculated Paving Speed - 95% Efficiency:	[19.3] *ft/min*	[5.89] *m/min*
Legal	Calculated Paving Speed - 90% Efficiency:	[20.2] *ft/min*	[6.17] *m/min*
	Calculated Paving Speed - 85% Efficiency:	[21.2] *ft/min*	[6.45] *m/min*
	Calculated Paving Speed - 80% Efficiency:	[22.1] *ft/min*	[6.73] *m/min*
	Calculated Paving Speed - 75% Efficiency:	[23.0] *ft/min*	[7.01] *m/min*
Exit	**Effective Paving Speed**:	[18.4] *ft/min*	[5.61] *m/min*

Because the paving width is narrower in Example Two, the paving speed increases to consume the tonnes delivered to the worksite each hour. The effective paving speed, or 100% efficiency factor, is 5.6 m/min (18.4'/min). At 75% efficiency, the actual paving speed increases to 7 m/min (23'/min).

Assuming that this is a binder (intermediate) course and that the wearing (surface) course will be thinner, calculate a speed for the wearing course shown in Example Three.

Example Three – Hourly production remains at 200 tonnes (220 tons) per hour. The weight of the uncompacted material is the same. Using the 3.66 m (12') paving width, enter 50 mm (2") paving thickness and calculate the paving speed.

PAVER SPEED CALCULATOR

Trucking	**General Inputs**	*Imperial Units*		*Metric Units*	
Paver Speed	Paving Thickness:	[1.97]	*in*	[50]	*mm*
Compaction	Paving Width:	[12.00]	*feet*	[3.658]	*meter*
Windrow	Material Density Uncompacted:	[135]	*lbs/ft³*	[2162]	*kg/m³*
Yield	**Paver Speed @ Given Production Rate**				
Slope	Production Rate of Hot Plant:	[220]	*tons/hr*	[200]	*tonnes/hr*
Thickness	Calculated Paving Speed - 100% Efficiency:	[27.6]	*ft/min*	[8.42]	*m/min*
Job Summary	Calculated Paving Speed - 95% Efficiency:	[29.0]	*ft/min*	[8.84]	*m/min*
Legal	Calculated Paving Speed - 90% Efficiency:	[30.4]	*ft/min*	[9.26]	*m/min*
	Calculated Paving Speed - 85% Efficiency:	[31.7]	*ft/min*	[9.68]	*m/min*
	Calculated Paving Speed - 80% Efficiency:	[33.1]	*ft/min*	[10.10]	*m/min*
	Calculated Paving Speed - 75% Efficiency:	[34.5]	*ft/min*	[10.53]	*m/min*
Exit	**Effective Paving Speed**:	[27.6]	*ft/min*	[8.42]	*m/min*

In Example Three, the paving thickness is less, so the paving speed increases to maintain the same production rate. As the paving speed begins to increase, three factors need to be considered.

1. Can the compaction process keep up with the paving speed? Notice that the Production Calculator includes a tab for calculating compactor working speed. Use that tab to verify that the correct compaction equipment will be available.

   To calculate compactor speed these factors must be known:

   - Drum width
   - Vibratory system frequency
   - Repeat passes to reach target density

The Paving Production Calculator verifies that a vibratory compactor with 170 cm (67") wide drums operating in the first compaction phase behind the paver and set in high frequency, 3 800 vibrations per minute, and making three repeat passes can match the paving speed.

The actual roller working speed is set at 89 m/min (292'/min). The pattern has a total of nine passes so the effective roller speed is 9 m/min (29'/min), a pace that exactly matches the paver's progress. Note that the efficiency factor is 85%. The efficiency factor compensates for water refill stops and non-vibratory, pattern-reverse times.

Pre-project planning should always include verification of the production match between the paver and the first phase compactor(s).

COMPACTION CALCULATOR

Trucking	Roller Model:	[*Click to Select Another Model*]	[**CB54**]	
Paver Speed	**General Inputs**	*Imperial Units*	*Metric Units*	
Compaction	Paving Width:	[**12.00**] *feet*	[**3.658**] *meter*	
Windrow	Actual Drum Width:	[**67**] *in*	[**170.18**] *cm*	
Yield	Amount of Overlap:	[**5.9**] *in*	[**15.0**] *cm*	
Slope	Speed of Vibrator:	[**3800**] *VPM*	[**3800**] *VPM*	
Thickness	Impacts (recommended):	[**13**] *per ft*	[**43**] *per m*	
Job Summary	**Number of Passes to Cover Mat Width Once:**		[**3**]	
Legal	Number of Repeated Passes (from test strips):		[**3**]	
	Total Number of Passes:		[**9**]	
	Roller Efficiency Rate (recommended 75 to 85%):		[**85**] %	

	Imperial Units	*Metric Units*	*Effective Paver Speed*
Actual Roller Speed:	[**292**] *fpm*	[**89**] *mpm*	[**27.6**] *ft/min*
Effective Roller Speed*:	[**28**] *fpm*	[**9**] *mpm*	[**8.42**] *m/min*

** Effective Roller Speed should be at least 100% but no more than 115% of the Effective Paver Speed.* [**% = 104**]

Exit

2. Will a vibrating and tamping screed be able to perform properly at the calculated speed? Screeds with tamper bars are usually limited to slower working speeds, when compared to screeds that employ only vibration. The tamper bar must deliver a certain amount of energy into the bituminous layer in order to create the relatively high screed-laid density that is required for some applications.

To deliver high energy and avoid creating ripples in the layer, the tamper bar needs to hit the surface between 1.5-4 times per linear centimeter. Depending on the frequency of the tamping system, the width of the tamping foot, and stiffness of the material, working speed may have to be adjusted to obtain the desired density and impact overlap.

3. If automatic grade and slope control are used, can the system react quickly enough to match the paving speed? High-speed paving—in excess of 18 m/min (60'/min)—is a difficult proposition. Mistakes happen quickly. And certainly, a tamping screed cannot function properly at that high speed.

However, there are times when laying down a relatively thin layer (less than 5 cm (2") the supply will be 225 tonnes (250 tons) per hour or more. Then paving speed will approach the limitations of automatic grade and slope control. For high-speed paving—any application in excess of 18 m/min (60'/min)—Caterpillar recommends paving that layer manually, without use of grade and slope control systems.

[SUMMARY]

Do not overlook the importance of pre-project planning, whether creating a formal project plan or simply creating a checklist. Once approved, be sure to communicate the plan to the appropriate personnel. Delegate authority to supervisors and crewmembers, so they can prepare for their responsibilities. Have a method in place to verify that all elements of the plan are in place before work begins.

Do not put off planning until the last minute. Good planning will prevent many time-consuming and costly errors that are common on poorly planned paving projects.


CAT
AP1055F
GRADE CONTROL
SE10



Unit 3
FUNDAMENTALS OF PAVING

Each paving crew member should understand how the paver works, and the basic theories of paving. That knowledge helps with quality control and on-the-job problem solving.



Proper fundamentals are one of the four principles of quality paving. When the crew takes shortcuts or loses focus on fundamentals, defects invariably appear in the layer of bituminous material.

First, crewmembers must know how to correctly use all paver controls. That means the crew must study and refer to the Operation and Maintenance Manual that is shipped with the paver. Controls and features vary on different paver models. A procedure or technique used on one paver may not be appropriate for another. Always set up and operate the paver in the manner described in the manufacturer's manual.

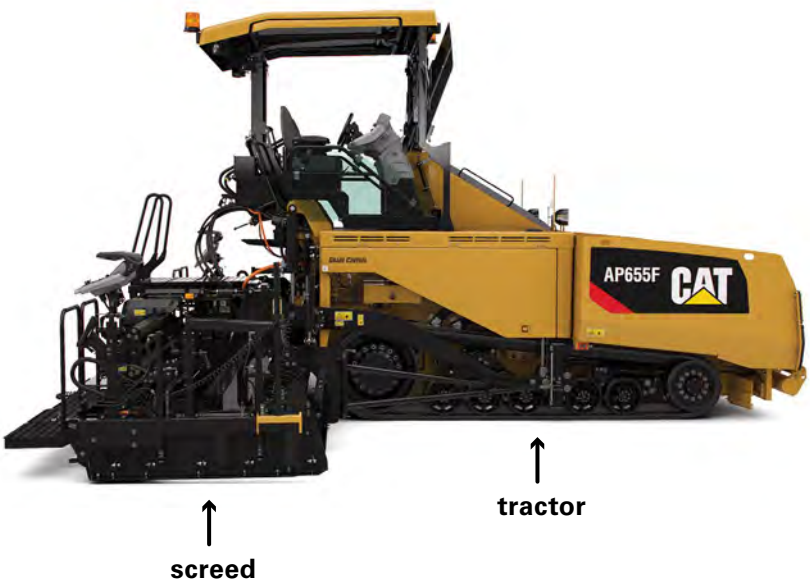
Second, crewmembers have to fully understand the concept of the free-floating screed. This concept is basically the same for all pavers. The material supports the screed, which is towed by the tractor. When all the factors that influence how the screed floats on the material are properly controlled and constant, then the screed is said to be at equilibrium and will maintain its vertical position.

Finally, the crewmembers must know all factors that influence how the screed floats. These include paving speed, the material feed system, screed adjustments, temperature of the bituminous material, and type of bituminous material. The crew is expected to control the first three factors: speed, material feed system and screed adjustments. The crew does not control mix temperature or mix design, but the crew needs to know how to react to variations in those factors.

[HOW THE PAVER WORKS]

An paver has two basic elements, the tractor and the screed. The tractor provides power to tow the screed. The tractor receives the material from haul units or transfer devices and moves the material to the rear of the tractor, where it is placed in front of the screed. The tractor can be equipped with one or two operating stations having controls for steering, the propel system, and parts of the material feed system.

The screed is connected to both sides of the tractor at the tow-point connections. From the tow-points, tow arms extend back and connect to the front of the screed. The screed controls the width and depth of paving. It also provides the initial density and texture of the bituminous layer.



[TRACTOR]

Two push rollers are located at the front of the tractor. When the mix is transferred directly from the haul unit to the tractor hopper, the push rollers contact the rear wheels of the haul unit. The push rollers turn as the wheels turn. The truck driver must maintain light pressure on the brakes to prevent it from rolling away and creating a spill in front of the paver. (See Unit 7 for more information about the truck exchange procedure.)



Two push rollers contact the rear tires of the haul unit as the paver pushes the haul unit.

User Tip: *Periodically clean the push rollers to avoid material build-up. If build-up occurs, irregular impacts between the paver and haul unit can lead to mat defects.*

A truck hitch is an option on many pavers. It provides a solid connection between the paver and the haul unit. The paver operator controls the truck hitch with a console switch. When the truck hitch is open, the truck backs up and stops just before contacting the push rollers. The paver moves forward slightly and contacts the rear tires of the haul unit. The operator then closes the truck hitch and the rollers on the two truck-hitch arms contact the wheel rims.

When the load is fully discharged, the operator opens the truck hitch to permit the empty haul unit to depart. Truck hitches help avoid large spills in front of the paver that are caused when a truck inadvertently loses contact with the push rollers.

Material from haul units can be discharged directly into the paver hopper or can be moved into the hopper using a transfer device. The hopper provides a limited amount of storage for bituminous material, so the paver can continue to operate for some period of time between haul units.

Flashing at the front of the hopper helps retain material. Hopper inserts are commonly installed when the crew is using a material transfer device. The hopper insert increases the storage capacity of the paver and makes it easier to pave continuously at normal paving speeds without stops between haul units.



The optional truck hitch helps provide a secure connection between the front of the paver and the haul unit.

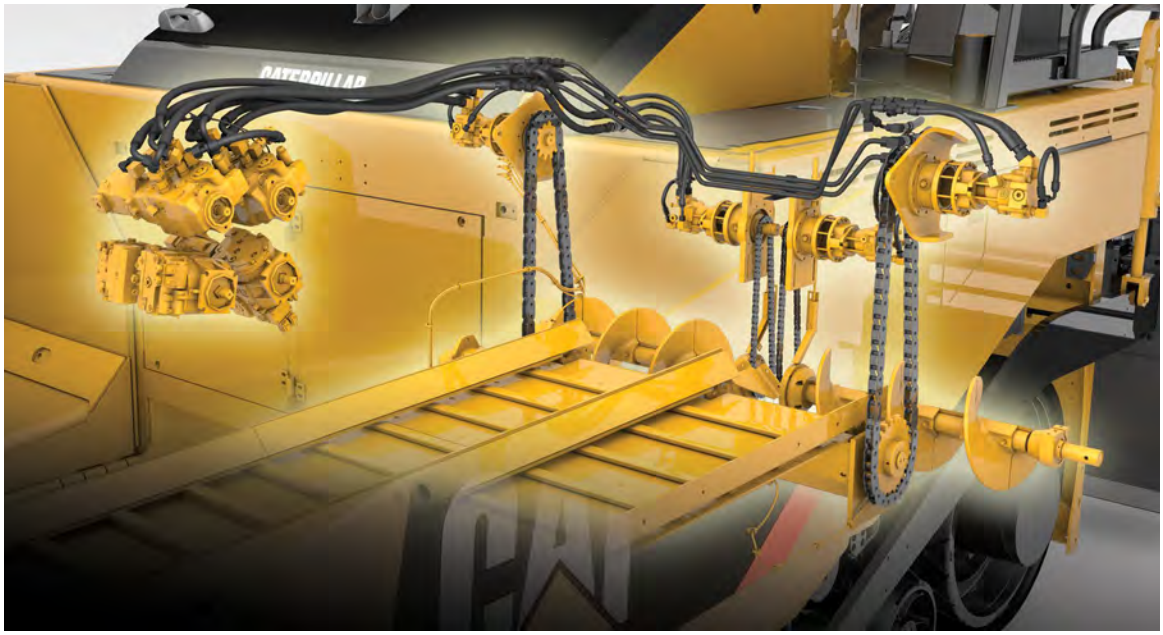
The sides of the hopper can be raised one at a time or simultaneously to use the material that has flowed to the sides. Local regulations sometimes prohibit raising hopper sides because doing so can cause material segregation.

Some pavers are equipped with an optional fold-up section at the front of the hopper. Folding helps prevent spills over the flashing when the sides of the hopper are raised.

Inside the hopper, two drag-slat conveyors move the material from the hopper, through tunnels under the operator's platform, and deposit the material in front of the screed. On modern pavers, the left and right conveyors are driven independently to meet the demands for material in front of the screed. For example, if the left side of the screed is set at a wider width than the right side of the screed, then the left feed system conveyor will run faster than the right conveyor.



The operator can raise the sides of the hopper, if desired and permissible, to move mix from the sides of the hopper to the feed system conveyors.



Left and right feed system conveyors move material to the front of the screed. Left and right augers move material out to the full paving width.

At the rear of the tractor, two augers move the material in the auger chamber out to the paving width. The left and right augers also feature individual controls to regulate their rotation speeds.

The material feed system has several types of sensors to help automatically regulate the material feed system. Either sonic or mechanical feeder sensors are located at the rear of the left and right conveyors and/or near the ends of the left and right augers.

The tractor can be equipped with one or two console equipped operating stations. The controls regulate propel speed, tractor steering, some material feed system controls, lights and some screed controls. The operator selects the station that provides the best visibility for steering or slides the station to the side of the tractor with the best visibility.

The paver operator's primary responsibilities are to help coordinate transfer of material to the hopper, guide the machine, and control speed. The operator is also responsible for setting some feed system controls.

Tractor undercarriages include wheel, rubber track, and steel track designs.

Before the introduction of the rubber track undercarriage, there was clear distinction in applications for the track-type and rubber-tire undercarriages.

Track-type systems were preferred for extra traction and flotation on soft base materials. Rubber-tire systems were preferred for worksite mobility and for projects where smoothness of the wearing course was very important. Rubber track undercarriage now combines traction, high travel speed and smoothness and is very common in some regions today.

Tow-point connections are located between the screed and the tractor near the center of the tractor on both sides.

A hydraulic cylinder is attached to the tow arm. The front of the tow arm extends through a vertical slide compartment. At the end of the tow arm is a roller that contacts the slide compartment. The screed or paver operators can adjust the height of the tow-point connection to meet application requirements. A scale shows the operators the height of the tow-point. The tow-points are the only physical connections between the screed and the tractor and they play an important role in how the screed floats.



The tow-point connection features a height scale that enables the tractor or screed operators to set and check the position of the connection.

[SCREED]

Screeds are classified into two primary types—fixed-width and hydraulically extendable, each having several variations.

Extendable screeds can have extenders that are in front of the main screed (called a front-mount screed) or are behind the main screed (called a rear-mount screed). There are vibratory screeds and tamping screeds with vibration.

Each screed has its own relative merits that fit certain applications better than others. In some areas, there may be method specifications that mandate the use of a particular type of screed. This guide will not debate the benefits of one screed type over another, but will seek to explain how to best set up and use any type of screed.

A fixed-width screed consists of one screed frame and screed plate. Typical fixed-width screeds are 2.5 m (8') or 3 m (10') wide. To increase the paving width, the crew must add bolt-on screed extensions. Fixed-width screeds are preferred for some applications, because they tend to make it easier to create uniform layer density from edge to edge, and they tend to resist flexing when paving at wide widths.

A hydraulically extendable screed consists of a main frame with screed plate plus right and left extender frames with plates. Typical main screeds are 2.5 m (8') or 3 m (10') wide, just like the fixed-width screeds. To increase paving width beyond the basic width, the operators use control switches to move the extenders.

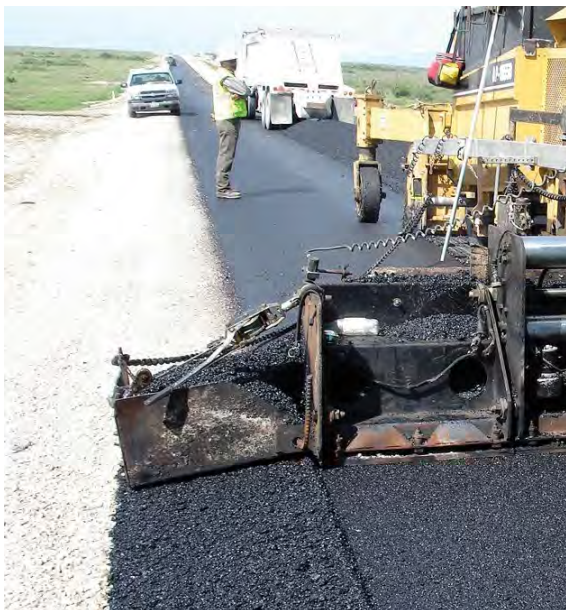


Left and right hydraulically extendable screed extenders simplify the task of changing paving width.

Depending on the particular screed arrangement, the total width of the screed can be increased to twice the width of the main screed by moving the extenders to their capacity. Those screeds are commonly referred to as "double-extending" screeds. Other hydraulically extendable screeds can be opened up to slightly less than twice the width of the main screed.

User Tip: *If possible, use equal-extension widths on both sides of the main screed, whether using bolt-on or hydraulic extenders. Equal-extension widths make it easier to control steering and balance material flow.*

Extendable screeds are nearly unlimited in their setup capabilities. Their versatility is why extendable screeds are most commonly used around the world. For example, an extender can be sloped to an angle that is different than the main screed plate when the application calls for paving an emergency lane with one slope, while paving the driving lane with another slope.



To meet the requirements of some applications, the crew must install bolt-on screed extensions and specialty attachments to the left and/or right screed extenders.

There are two types of extendable screeds. They include screeds with front-mount extenders and screeds with rear-mount extenders. The front-mount screed is equipped with extenders that move in and out in front of the main screed. In general, it is easier to retract front-mount extenders because the material brought in by the end gates is unrestricted. For this reason, front-mount screeds are often preferred by crews that pave urban-type projects where frequent width changes are required.



Front-mount screeds have extenders that move in and out in front of the main screed.

User Tip: *If using an extendable screed with front-mount extenders, and the paving width will constantly extend at least 60 cm (2') outside the main screed width, be sure to add auger and mainframe extensions. Remember, the material must flow straight out from the auger chamber to reach the end gates. Failure to add on auger and mainframe extensions will make it difficult for the crew to adjust the feeder system.*

Rear-mount screeds have extenders that are located behind the main screed. In general, rear-mount screeds are preferred by crews that pave mainline and highway projects, requiring fewer paving-width changes.

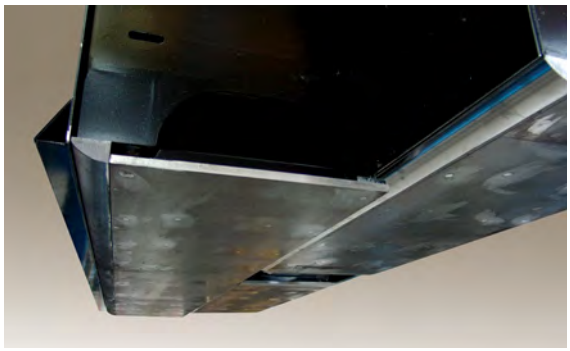
User Tip: *When using an extendable screed with rear-mount extenders and the paving width will constantly be at least 1 m (3') outside the main screed width, be sure to add auger and mainframe extensions. Rear-mount screeds provide a little leeway compared to front-mount screeds. The material exits the auger chamber and has time to flow out and back to fill in the area in front of the extenders. Failure to add on auger and mainframe extensions make it difficult to adjust the feeder system.*



Rear-mount extendable screeds have extenders that move in and out behind the main screed.

Vibratory screeds have small, external eccentric weight shafts installed over each screed plate. The rotation of the eccentric weights causes the screed plates to vibrate when the vibratory system is activated. Vibrating plates increase the density of the layer by an additional one or two percentage points.

For example, if a non-vibratory screed produces an asphalt layer that is at 84 percent of maximum theoretical density, that number will increase to 85-86 percent if the vibratory system is activated—all other factors being equal. Screed vibration also helps bring the fine mix particles to the surface and improves mat appearance.



View of the screed plates on a rear-mount, vibratory, extendable screed.

User Tip: *Adjust the frequency of the screed vibratory system while the paver is moving at normal operating speed. The goal is to deliver the maximum vibration energy into the bituminous layer without causing excessive vibration of screed components. A common technique is to place a coin or metal washer on any flat screed surface. Increase vibratory frequency until the coin begins to bounce. Reduce frequency until the coin just stops bouncing. If the paving speed is changed, adjust the vibratory frequency.*

Tamper screeds supplement vibration energy with the energy of one or two tamper bars. Tamper bars are located in front of the main screed and the screed extenders. The tamper bars are connected to an eccentric drive shaft that produces rapid vertical movement of the tamper bars. The movement helps material, especially stiff mixes with large aggregates, flow under the screed nose. The extra help means that the screed can float on the layer using a relatively small planing angle (angle of attack).



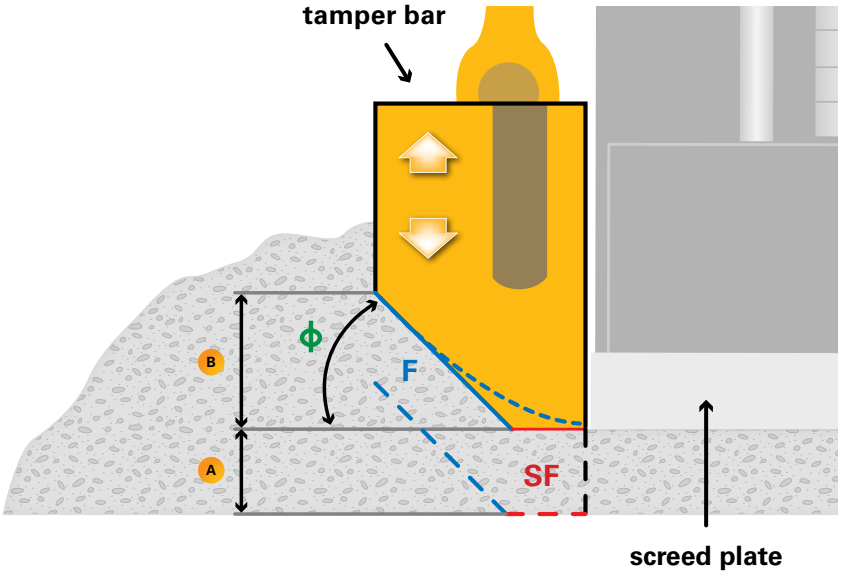
View of rear mount, extendable screed with vibration and tamper bars.

The distance the tamper bar moves and the frequency (rotations per minute) of movement depend on the screed model. For example, the tamper bar movement up and down may be 8 mm (0.3") at a maximum frequency of 2000 rotations per minute. The effect of the tamper bar energy will be to increase the density of the layer laid down by the paver beyond the results from vibration only.

For example, if the density of the layer is 86 percent of maximum theoretical density using only vibration, density may increase to 90 percent after activating the tamper bar system.

The working speed, the frequency of the tamper bar and the width of the tamper foot are factors that affect the results. The tamper foot (bottom of the tamper bar) is 25 mm (1") wide when new. The tamper foot must impact the surface of the layer in a closely spaced manner to create an overlap of about 6 mm (0.25") for each stroke of the tamper bar. In general, the maximum working speed is around 8 meters per minute (26' per minute) when paving with a tamper bar screed. Excessive working speed can cause ripples in the surface of the bituminous layer.

- A *tamper amplitude*
- B *dosing height*
- ϕ *dosing angle*
- F *dosing area*
- SF *tamper foot*



User Tip: *As the tamper bar accumulates use, the dosing angle changes and the area of the tamper foot becomes smaller. Therefore, the overlap between tamper strokes is harder to maintain. Regularly inspect the tamper foot and replace it when the wear affects smoothness or reduces production. As abrasion reduces the size of the tamper foot, the paving speed must be reduced to avoid creating ripples in the surface of the bituminous layer and to achieve the desired pre-compaction layer density.*

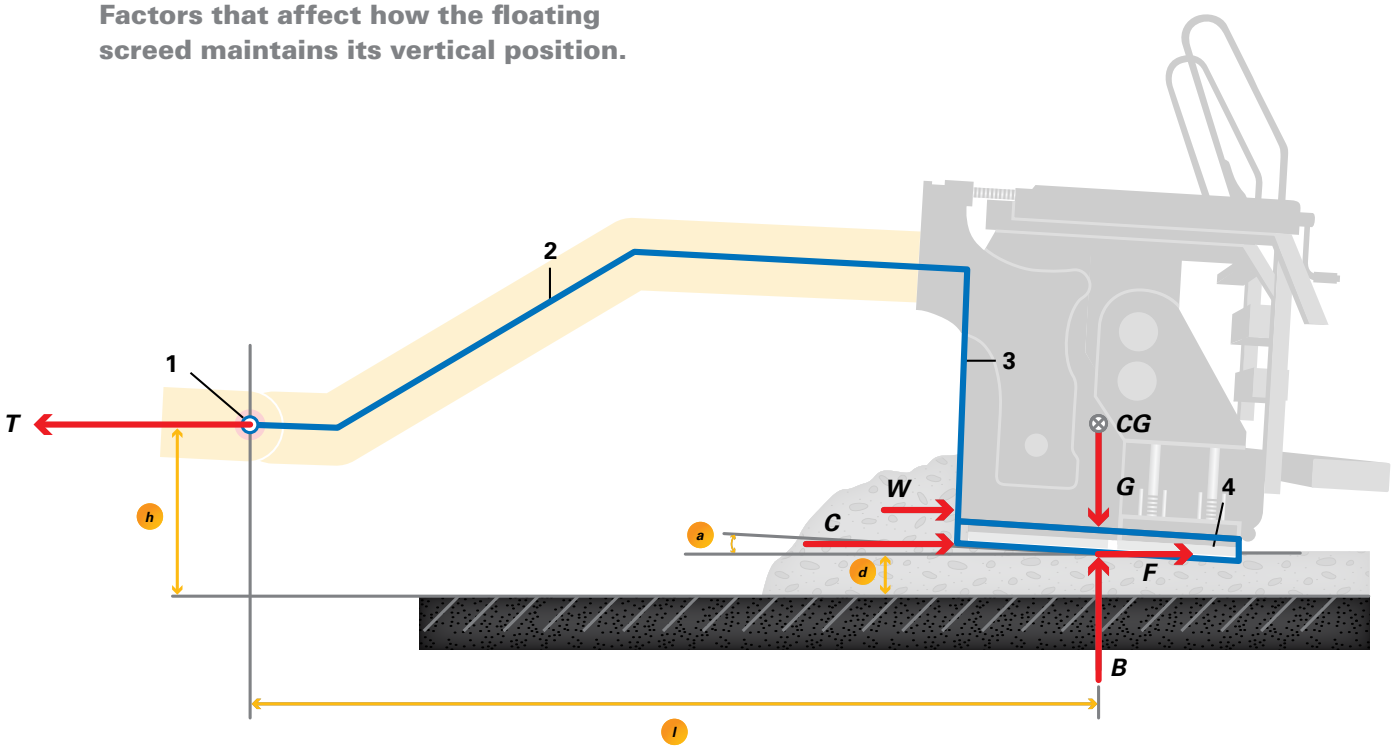
[PRINCIPLE OF THE FLOATING SCREED]

All screeds share one common principle: They float on the bituminous material that passes under the nose of the screed. When the paver operator lowers the screed to the starting reference, the operator positions the screed Raise / Lower / Float switch in the Float position. While the paver is moving, the bituminous material supports the screed.

Depending on the weight of the screed and the stiffness of the material, the operator may elect to put some upward hydraulic pressure in the screed lift cylinders to help the screed float at the proper angle. Also, the operator may choose to hold the screed in position during stops to prevent leaving indentations in the uncompacted layer. (More information about these systems is provided near the end of this unit.)

Both vibratory screeds and vibratory / tamping screeds float on the material. Their characteristics are somewhat different, but the factors that impact how they float on the material are basically the same. These factors are best explained in a diagram.

Factors that affect how the floating screed maintains its vertical position.



1 *tow-point connection (rotary joint)*
2 *tow arm (beam)*
3 *front panel*
4 *screed plate*
T *traction force*
W *weight of material in front of the screed*
C *cutting force (shear force)*
G *gravitational pull (weight of the screed)*
B *buoyant force*
F *frictional force*

h *height of the tow-point*
l *projected length*
d *depth of the layer (mat thickness)*
a *angle of attack*

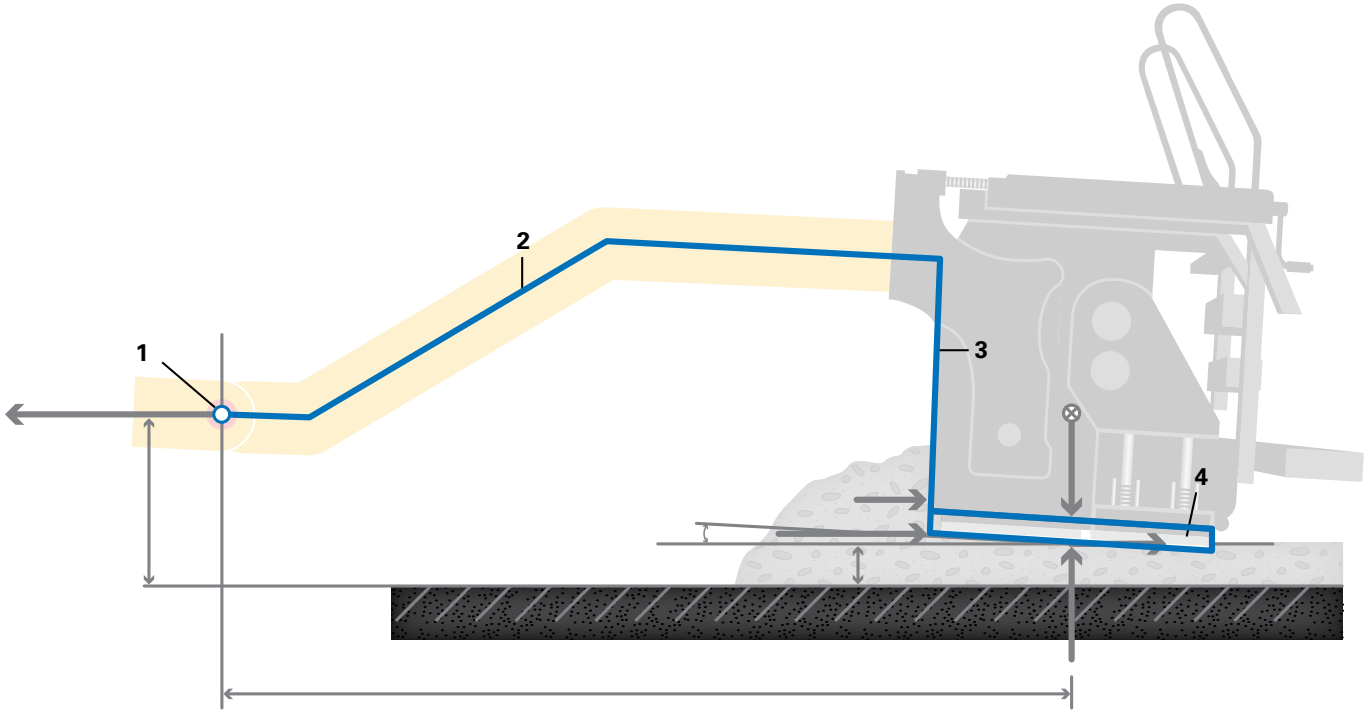
First, let's identify the main components in the diagram.

Number 1 is the tow-point connection (rotary joint). There are connections on each side of the paver. The height of the tow-point connections is adjustable. (See Units 4 and 5 for information about setting the tow-point height.) Changing the height of a tow-point connection affects the main screed angle of attack and thereby the vertical position of the screed.

Number 2 is the tow arm (beam) that extends between the tow-point connection and the front of the screed. There are identical tow arms on each side of the paver.

Number 3 is the front panel of the screed. This panel confines the material in the auger chamber. A certain amount of material is pushed in front of the panel, while a certain amount flows under the front panel.

Number 4 is the screed plate, which floats on material that is sheared off by the front panel. In conjunction with the tamper bar (if equipped) and screed vibration (if engaged), the screed plate develops an additional amount of mat density as the mat is being laid down. The screed plate also affects the texture of the mat surface.



1 *tow-point connection (rotary joint)*

2 *tow arm (beam)*

3 *front panel*

4 *screed plate*

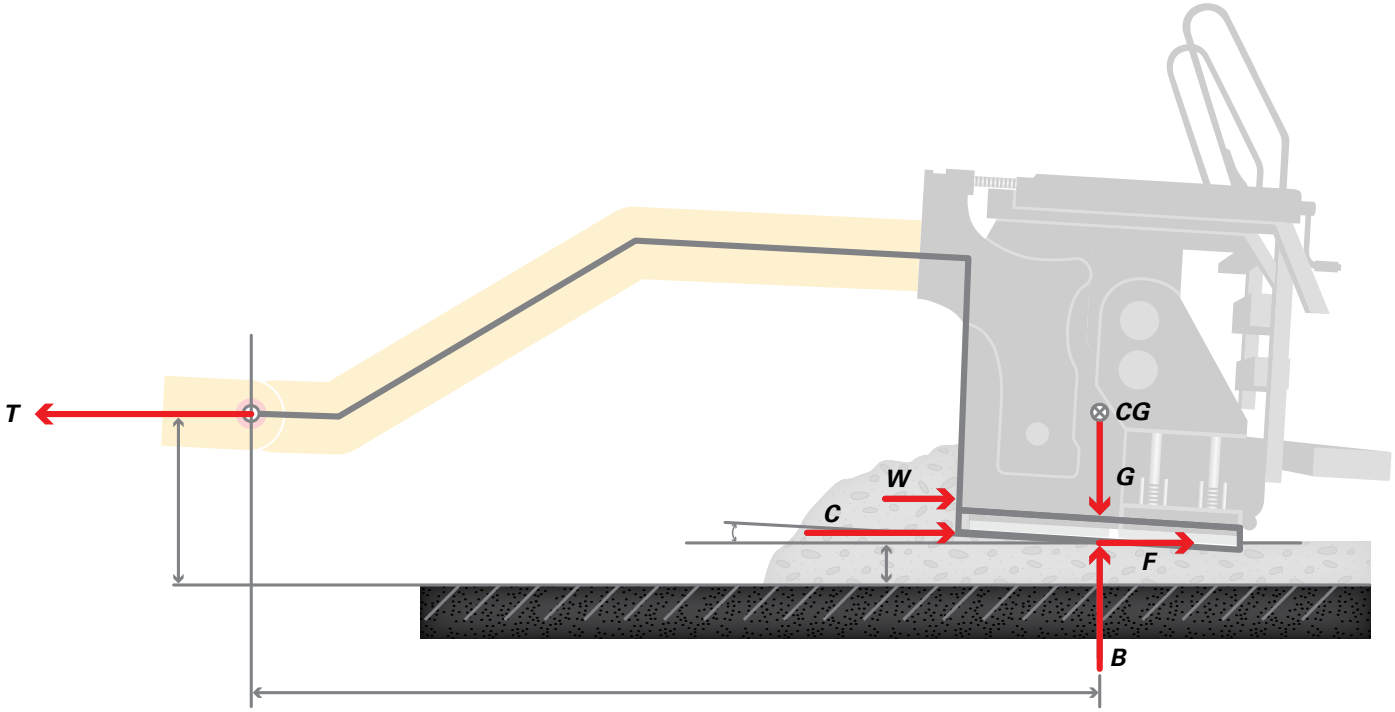
Next, are the forces that act on the screed.

Force *T*: traction force. The traction force is controlled primarily by the speed of the paver. The paving speed provides the energy that enables the screed to shear a portion of the material in the auger chamber. Uniform paving speed contributes to screed equilibrium and, therefore, a uniform layer thickness and smoothness.

Force *W*: weight of material in front of the screed. This weight, or force, creates resistance. Weight is affected primarily by the material height in front of the screed, also called the head of material. The higher the head of material, the heavier the weight. The heavier the weight, the greater the resistance felt by the screed—and the more the screed wants to climb onto the head of material.

Conversely, the lower the head of material, the lower the resistance, and the screed pushes more material in front while allowing less to pass under the front of the screed. When the head of material remains uniform, the screed is more likely to float at equilibrium.

Resistance felt by the screed also depends on the action of the augers. When the augers rotate very slowly or stop rotating altogether, the material compresses against the front of the screed and its weight increases. Consequently, the screed will tend to ride up and thickness increases. Uniform auger rotation in the range of 20 to 40 revolutions per minute helps the screed maintain equilibrium.



T *traction force*

W *weight of material in front of the screed*

C *cutting force (shear force)*

G *gravitational pull (weight of the screed)*

B *buoyant force*

F *frictional force*

Note for tamper screeds: *If the paving speed is increased and the tamper and vibration speed remains constant, the distance between tamper impacts increases. Less material will be consolidated under the screed and the thickness will decrease. To maintain the correct layer thickness at higher working speeds, tamper and vibration speeds must be increased accordingly.*

The opposite effect occurs when working speed is decreased. Impacts delivered by the tamper bar are closer together and more mix passes under the screed. To restore the balance between working speed and tamper forces, the speed of the tamper bar will have to be decreased.

Force *C* : cutting force (shear force). The cutting or shear force is the screed's ability to pass a certain amount of mix under the front of the screed, while pushing the remainder of the material in the auger chamber forward. The shear force is affected by paving speed, head of material, resistance of material influenced by rotation of the auger, and angle of attack.

Force *G* : gravitational pull (weight of the screed). The weight of the screed acts to compress the layer as material passes under the screed. If all other factors are equal, a heavier screed wants to compact the material to a greater extent. The crew members may have to adjust to make a heavy screed reach a stable equilibrium point. If the width of the screed remains unchanged, the weight force remains nearly the same.

If the width of the screed is changed significantly while paving, the force changes and the crew will have to make adjustments to maintain the correct depth. For example, if extenders are brought out, the weight felt by the material is less and the screed will rise.

Force *B* : buoyant force. This is the hydrodynamic lifting force that is created by the material passing under the screed. Mix designs that contain large aggregates (19 mm / 0.75" or greater) and high viscosity with modified asphalt cement tend to have larger lifting forces than fine-grained mixes with conventional asphalt cement. This lifting, or supporting force, remains nearly constant once the screed has reached equilibrium and the thickness of the bituminous layer and the mix design remain the same.

Force *F* : frictional force. The frictional force that is created between the screed plate and the mix passing under the screed plate is generally fairly small. As long as the mix formula and temperature remain the same, the frictional force stays constant and screed equilibrium is not affected.

Colder-than-normal mix tends to create less drag, or friction, on the screed plate. Under this condition, the screed will tend to fall. The larger the temperature loss in the mix, the more the screed will fall.

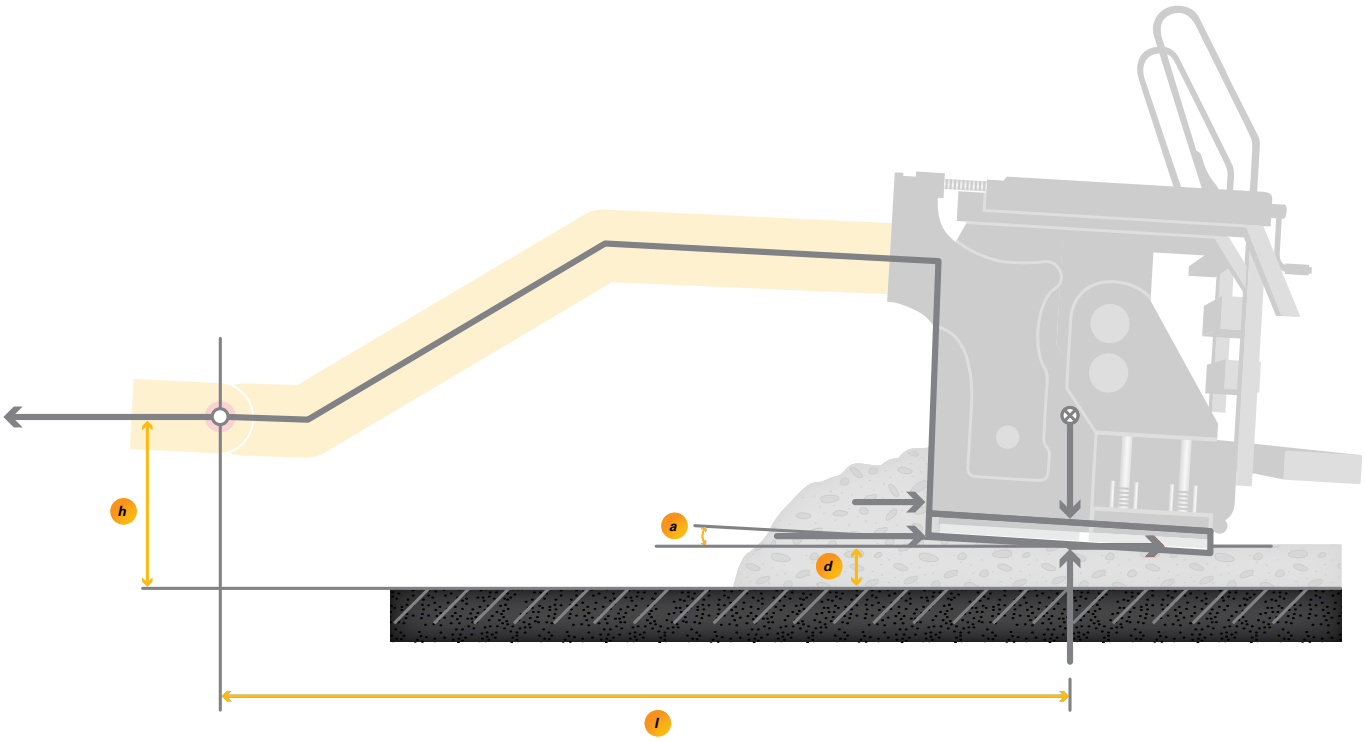
Next, let's examine the dimensional characteristics in the diagram.

Dimension *h* : height of the tow-point. The height of the tow-point connection affects the traction angle, or line of pull. The traction angle is the relationship between the height of the tow-point connection and the pivot point of the screed.

Before starting to pave, position both tow-point connections at the height that is appropriate for the planned layer thickness. (See Unit 4 for more information about setting the tow-point height for vibratory screeds. See Unit 5 for more information about setting the tow-point height for tamper screeds.)

The traction angle that is formed by the height of the tow-point connection affects the main screed angle of attack. On tamper screeds, this is the normal way to change the angle of attack. On vibratory screeds, crews use either the tow-point or manual depth cranks to change the angle of attack.

Increasing the traction angle increases the angle of attack and the screed will climb. Decreasing the traction angle decreases the angle of attack and the screed will go down. Screed operators can change the height of the tow-point connection manually using control switches or automatically using a grade control system.



- *h* — *height of the tow-point*
- *l* — *projected length*
- *d* — *depth of the layer (mat thickness)*
- *a* — *angle of attack*

User Note: *When paving manually (not using automatic grade and slope controls), the leveling characteristic of the free-floating screed can produce up to 50 percent improvement in smoothness when paving over a moderately to severely rough grade. The screed at equilibrium tends to remain at a given height, while filling in low spots in the grade and leveling off high spots.*

Dimension *l*: projected length. The projected length of the paver is the distance between the tow-point connection and the screed's center of gravity (indicated as CG on the page 70 diagram), or the screed pivot point. The projected length is often referred to as the paver's leveling distance. When the paver is moving and the screed is at equilibrium, the leveling distance creates a smooth surface—even when paving over an irregular grade.

Dimension *d*: depth of the layer (mat thickness). Layer depth is the distance between the bottom of the screed plate and the grade. Layer depth affects where the crew should position the tow-point in order to create the ideal traction angle (line of pull) and the ideal angle of attack for the main screed. Specifically, when the paving depth is 15 cm (6") or greater, the crew will have to supplement the traction angle with increased angle of attack to help the screed float in a stable condition.

Dimension *a*: angle of attack. The angle of attack is also referred to as the planing angle. Angle of attack can be thought of in several ways. It can be described as the angle between the nose of the screed plate and the grade over which the screed is paving. Or, it can be described as the relationship between the nose and the trailing edge of the screed plate. In general, the best screed stability and best appearance of the bituminous layer occur when the angle of attack is between 3 mm (0.125") and 6 mm (0.25").

The primary influence on main screed angle of attack for tamping screeds is the height of the tow-point connection. The angle of attack on vibratory screeds can be set or altered in two ways: turning the depth crank or changing the height of the tow-point connection.

Summary. The forces that affect the screed are interrelated. If one factor is changed, one or more other factors may need adjusting to create the conditions necessary for screed equilibrium. Therefore, the crew should strive for uniform performance at all times.

However, even with the best planning and preparation, it is virtually impossible to keep all forces constant. Setting up the screed properly can minimize the negative effects of fluctuations in the forces that affect the screed. Pay attention to the traction angle (line of pull) and the angle of attack that is created at the start of paving. Having the screed in the most stable condition will help reduce the screed's reactions to changing forces.